

AI4Mobility

A multimodal assistive platform for parents supporting children with mobility difficulties. This deck covers how the body is drawn, explored and compared against a reference position.

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What this deck covers

From one connection list to a live camera coach, with the caveats attached.

1

Drawing the body

The skeleton connection list, the 46-pose illustration library, and the coordinate bug that shaped both.

2

Exploring the data

Angle distributions, the PCA projection, and what 52.3% of variance does and does not prove.

3

The live pose coach

On-device pose estimation, four joint checks, and where the target angles come from.

4

Results and limits

The model comparison, plus an honest account of what is not validated.

Team Deepminds — four roles, one platform

Four members, four defined responsibilities. This deck presents the pose estimation and visualisation work.

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Data engineering and pipeline lead

Peer contribution split: 28%

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Model development and evaluation lead

Peer contribution split: 26%

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THIS DECK

Pose estimation and visualisation lead

Peer contribution split: 24%

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Research and documentation lead

Peer contribution split: 22%

01

Drawing the body

One connection list, 46 hand-authored poses, and the coordinate convention that had to be right before anything else could be.

One connection list, used everywhere

Thirteen landmarks, joined the same way in the notebook figures and in the platform illustrations.

The connections

- Head to neck, then neck to each shoulder.
- Shoulder to elbow to wrist, on each side.
- Neck to hip as the trunk segment.
- Hip to knee to ankle, on each side.
- The platform renderer extends the same list to feet and hands so the illustrations read clearly.

Why it is shared

- A figure in the report and a figure in the interface describe the same body.
- The classifier features are computed at the same joints the drawing connects.
- The coach checks angles at vertices that already exist in the drawing.
- One definition means one place to correct if a joint is wrong.

46 hand-authored stick-figure poses

- 1 Every step in every programme has a picture**
Poses cover lying, kneeling, quadruped, seated and standing positions across the ten programmes.
- 2 Props are drawn where the exercise needs them**
Chair, wall, step and table, so the picture matches the instruction rather than a generic figure.
- 3 Drawn as inline SVG in the browser**
No image files to load, so the illustrations scale with the accessibility settings and reduced-motion preference.
- 4 Hand-authored joint coordinates**
They are not the output of a pose estimation model, and the code says so in its own header comment.

WHAT THIS STANDS IN FOR

Generative illustration is NOT built. The 46 poses are a hand-authored stand-in for a generative vision model, and the register in the report records it as such.

FIGURE 1

The same skeleton, drawn from the data

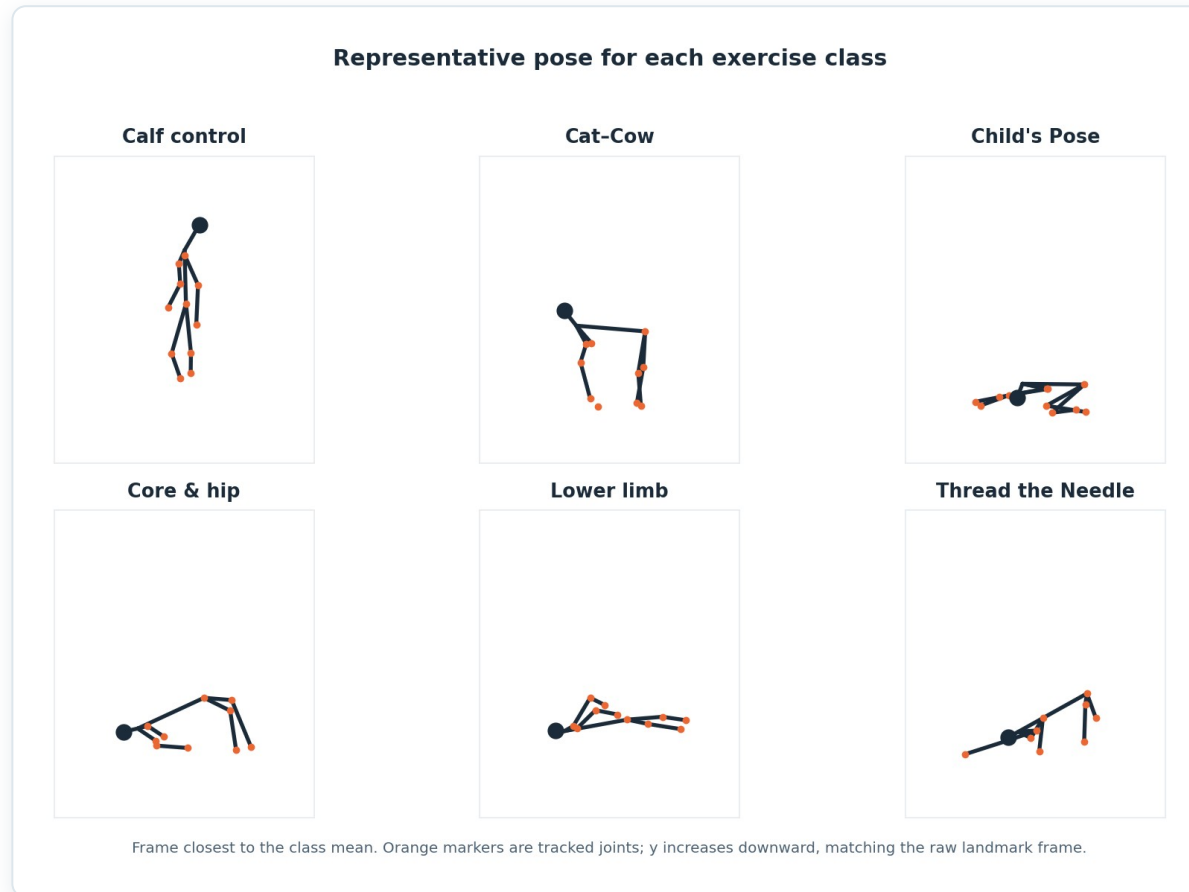


Figure 1 — one representative frame per class, drawn with the shared connection list.

How to read it

- Each panel is the frame closest to that class mean, not the first frame available.
- Orange markers are tracked joints; the larger dark marker is the head.
- y increases downward, matching the raw landmark frame rather than a mathematical axis.
- The quadruped-derived classes look alike here, and the confusion analysis later says the same thing numerically.

The image-coordinate inversion problem

Landmark data uses the image convention, where y increases downward. Plotting libraries do not.

The symptom, and how it surfaced

- Drawn on a standard axis, every pose came out vertically mirrored.
- It was caught in visual review of the pose gallery: the standing class rendered head-down against the floor.
- The lying and quadruped classes looked merely unusual, not obviously wrong — the failure was silent for four of six classes.
- No metric would have caught it: the features are computed from coordinates, not from the picture.

The fix, and what it changed

- The plotting axis is inverted so the drawing matches the image frame.
- The same convention is stated in the generator, the notebook renderer and the platform renderer.
- The platform draws in a viewBox where y increases downward, with the floor at a fixed line.
- Every later figure inherits the convention rather than repeating the correction.

02

Exploring the data

Angle distributions, a two-component projection and a confusion matrix — read carefully, not optimistically.

The dataset is synthetic

It is not recorded video of children and it is not a public dataset. Every figure in this deck describes the generator.

What the figures license us to claim

- The visualisation pipeline works and is reproducible under `random_state=42`.
- The classes are separable in the way the generator constructed them to be.
- The exploratory findings agree with the model results, which is a consistency check on the method.
- The same figures would be produced, unchanged, from a recorded dataset.

What they do not license

- No claim that a real child's pose would separate this cleanly.
- No claim that the live coach has been validated on anyone.
- No clinical claim about posture or technique.
- No claim that the illustrations have been reviewed by a physiotherapist.

FIGURE 4

Angles separate some classes far better than others

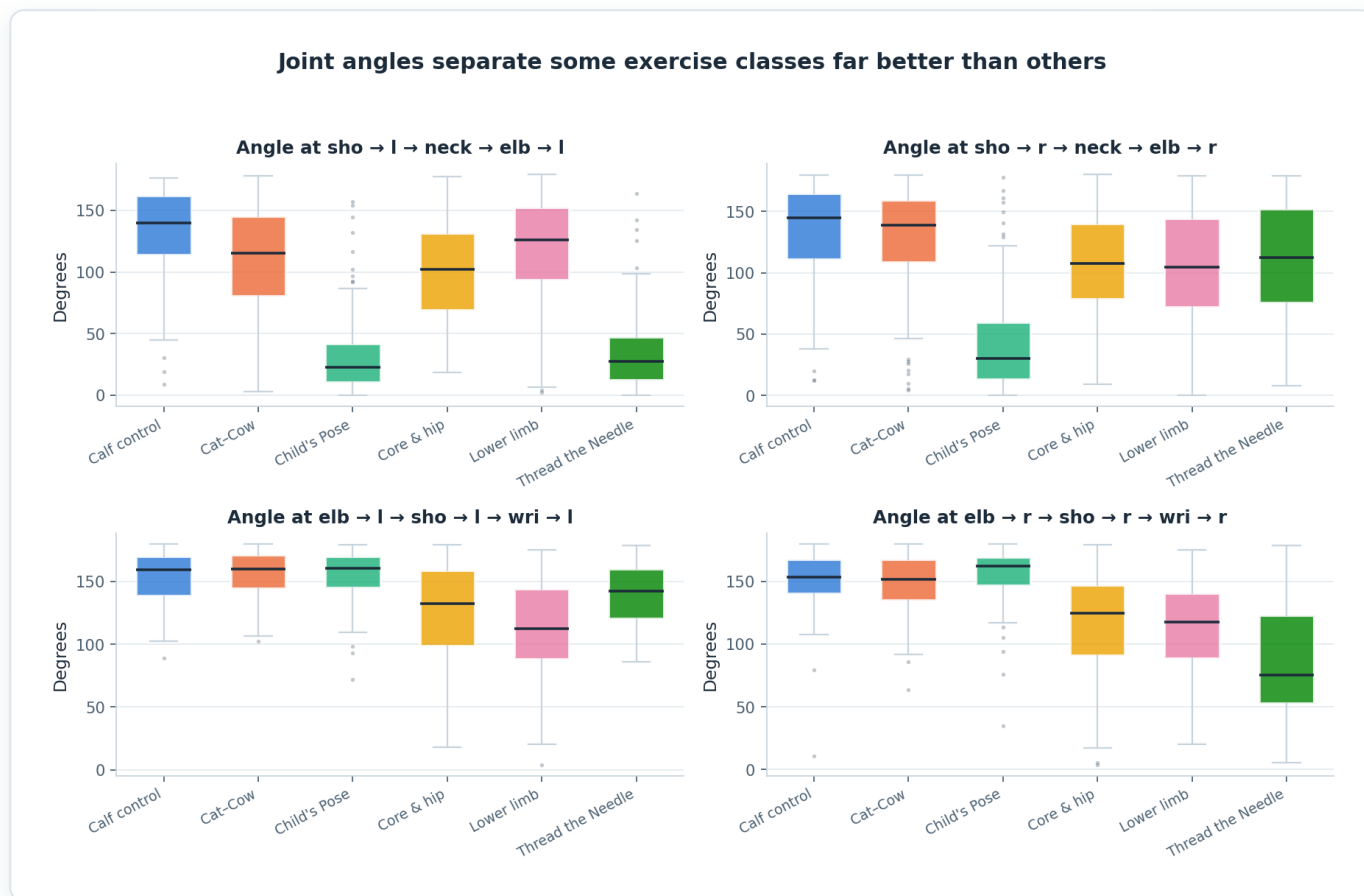


Figure 4 — joint-angle distributions by class.

What the boxplots say

- Elbow and shoulder angles separate the standing and quadruped groups clearly.
- Within the quadruped group the distributions overlap heavily.
- Angles are recorded as missing rather than zero when a segment is degenerate.
- This is the first place the later confusion structure becomes visible.

KEEP THIS IN PROPORTION

Angles are informative but they do not dominate the model: normalised positions carry 77% of feature importance.

FIGURE 5

PCA: 52.3% of variance in two components

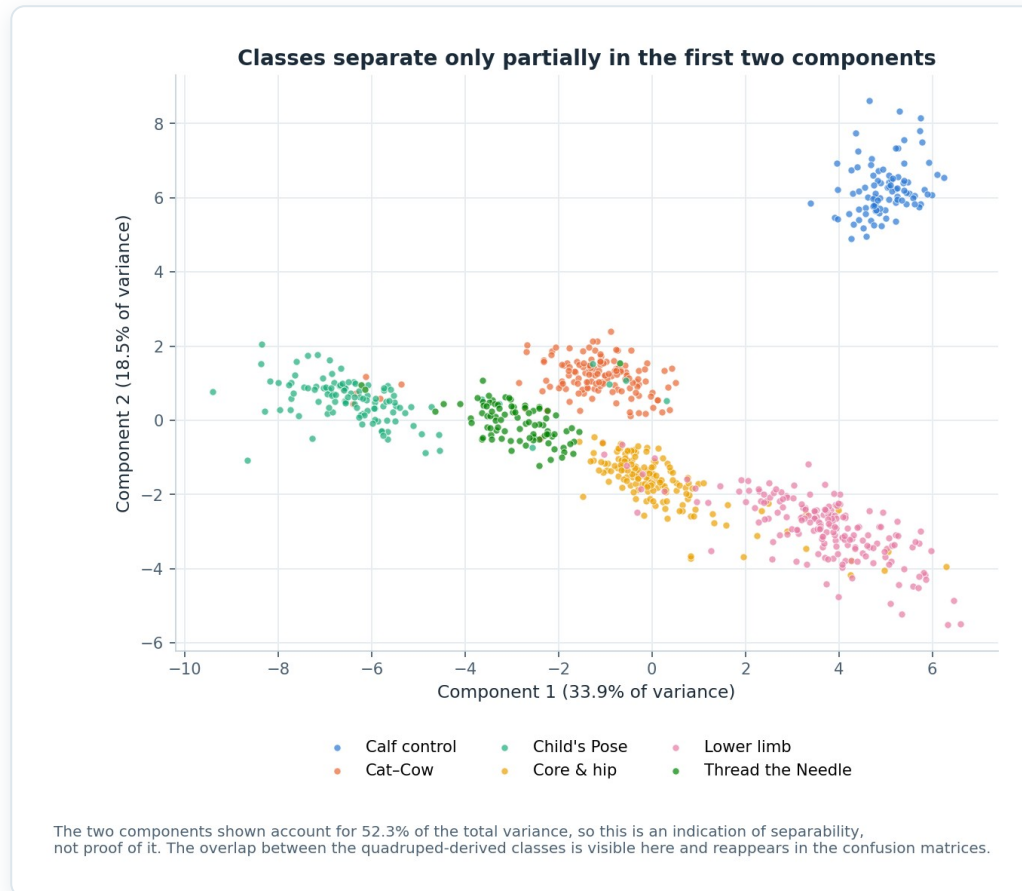


Figure 5 — PCA projection of the feature table, coloured by class.

Reading it correctly

- Two components account for 52.3% of total variance.
- Classes form visible groups, and the groups that touch are the ones performed in similar body positions.
- Just under half the variance is not shown at all in this view.
- A projection that separates classes is encouraging; it is not a measurement of accuracy.

AN INDICATION, NOT PROOF

This figure indicates separability. It does not prove it. The evidence for separability is the held-out test score, not the projection.

FIGURE 8

The confusions land where the projection said they would

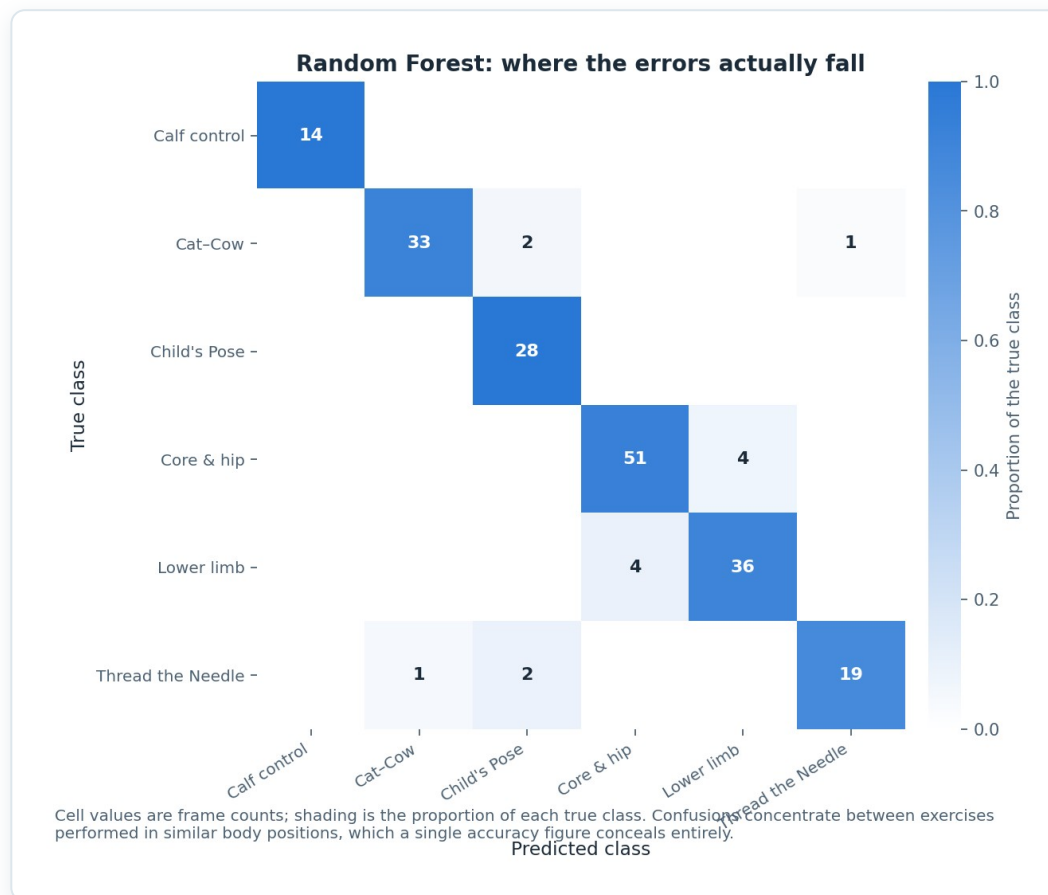


Figure 8 — Random Forest confusion matrix on the held-out test set.

The pattern

- Errors concentrate between exercises performed in similar body positions.
- Quadruped-derived classes are confused with each other; the two lying-down classes with each other.
- lower_limb and thread_the_needle are the weakest classes across all three models, at F1 near 0.90.
- The structure was built into the generator deliberately, so finding it is a check that the pipeline measures what we intended.

03

The live pose coach

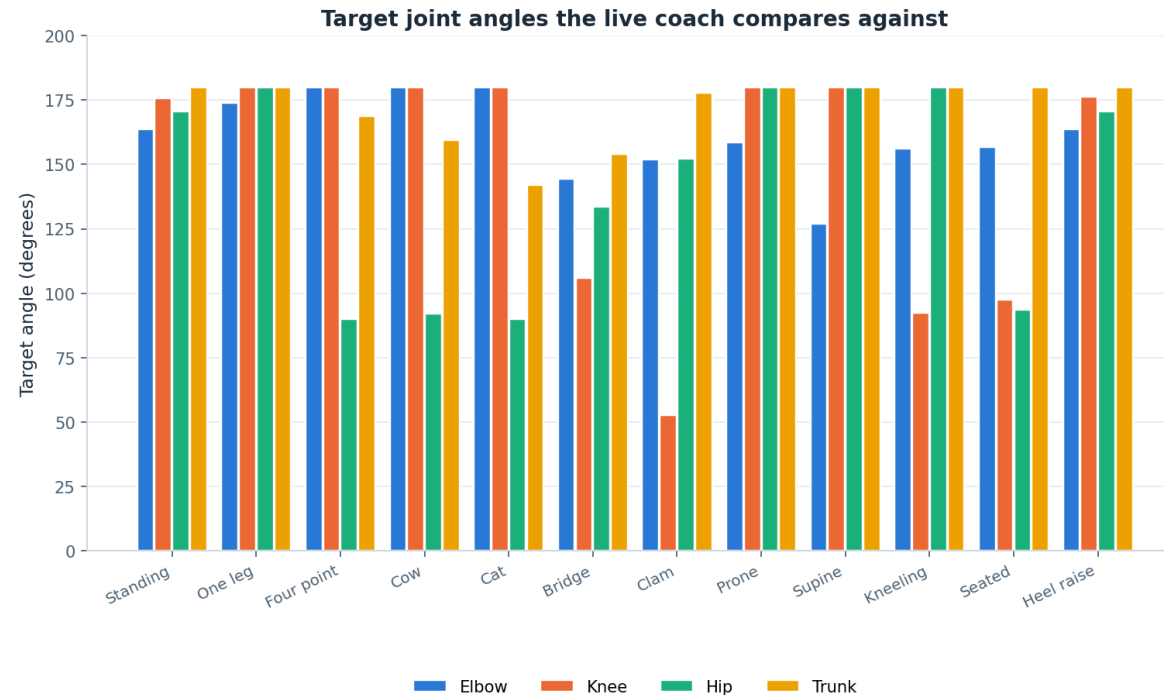
On-device pose estimation, four joint checks, and targets taken from the same pose bank the illustrations use.

Four joints, checked against the picture on screen

- 1 MediaPipe Pose Landmarker, running on-device**
The video element is the only place the image exists. No frame is uploaded and no recording is made.
- 2 A mirrored skeleton overlay**
Drawn on a canvas over the video, so the child sees themselves the way a mirror would show them.
- 3 Elbow, knee, hip and trunk angles**
Each is compared against a target for the pose the parent has selected.
- 4 One plain-English cue, plus the detail**
A single cue to act on, with a per-joint measured-against-target breakdown underneath for the parent.
- 5 It degrades cleanly**
If the model cannot be fetched, the browser has no camera, or permission is refused, the coach says so and the rest of the interface keeps working.

FIGURE 12

Target joint angles per pose family



Derived from the same hand-authored pose bank the illustrations are drawn from, so the coach checks the child against exactly the position the picture shows. Tolerances are 20-24° per joint and have not been clinically validated.

Where the targets come from

- Each target is computed from the joint coordinates of the pose the illustration draws.
- Different pose families give genuinely different target profiles, which is why one global threshold would not work.
- The trunk and hip targets carry most of the difference between lying, kneeling and standing families.
- The figure is drawn from the same bank the interface renders, so it cannot drift out of step.

Figure 12 — target joint angles by pose family, derived from the hand-authored pose bank.

The coach checks the position the picture shows

Target angles are derived from the same hand-authored pose bank the illustrations are drawn from.

How the derivation works

- The pose bank stores joint coordinates for every illustrated pose.
- For a selected pose, the angle at each checked vertex is computed from those coordinates.
- Pose aliases are followed, so a flow position resolves to the pose it is drawn as.
- Results are cached per pose, so the derivation happens once rather than every frame.

Why it matters

- The child is checked against exactly the position the picture shows them.
- There is no second set of numbers to maintain or to fall out of step.
- Correcting an illustration corrects the coach at the same time.
- It also bounds the claim: the coach is only as right as the hand-authored pose.

The tolerances are engineering judgement

Twenty to twenty-four degrees per joint, chosen so the coach is usable — not derived from clinical evidence.

- **Tolerances of 20 to 24 degrees per joint, set per checked joint.**
Tighter values made the coach reject positions a parent would call correct; looser values accepted positions that were visibly wrong.
- **That is a usability judgement, not a clinical threshold.**
No physiotherapist has reviewed these numbers, and no calibration study has been run.
- **The coach reports measured against target rather than pass or fail alone.**
A parent can see how far off a joint is, instead of being told only that something is wrong.

RECORDED AS NOT VALIDATED

Pose coach calibration is listed as NOT VALIDATED in the project register. The next step is a physiotherapist review of both the tolerances and the pose bank they are derived from.

Where the visual findings meet the model results

Three model families, identical data and identical splits. Measured on synthetic data.

Model	Test accuracy	Test macro F1	CV macro F1 (5-fold, train only)	Train accuracy	Generalisation gap
Random Forest	0.928	0.935	0.955 ± 0.017	0.965	+0.037
Gradient Boosting	0.923	0.931	0.939 ± 0.016	1.000	+0.077
Neural Network (MLP)	0.923	0.932	0.939 ± 0.019	1.000	+0.077

Cross-validation is five-fold on the training partition only. The test set was scored once, after all model selection was complete.

THE VISUAL AGREEMENT

Random Forest leads on macro F1 at 0.935 with the smallest generalisation gap at +0.037. Its weakest classes are the ones the PCA projection and the angle boxplots both flagged.

What the visual work does not establish

- **The illustrations are hand-authored, not generated.**
Generative illustration is not built; 46 poses are a stand-in for it.
- **No physiotherapist has reviewed the poses or the tolerances.**
Clinical validation is not done, and neither the pose bank nor the coach thresholds have been checked by a clinician.
- **The coach has not been used by a child.**
User evaluation is not done. Every judgement about usability is the team's own.
- **PCA shows 52.3% of variance, not the whole picture.**
Just under half the variance is outside the projection, so the separation shown is partial.
- **Everything visual rests on synthetic frames.**
The figures describe the generator, and would need redrawing from recorded data before any external claim.

What would strengthen this work

Ordered by how much each one would change what we are allowed to say.

1

Have a physiotherapist review the pose bank and the tolerances

This converts engineering judgement into a reviewed reference, and it improves the coach and the illustrations at once.

2

Record a small consented dataset

Redraw every exploratory figure from real landmarks and see whether the class structure survives.

3

Add temporal features and re-run the projection

Movement over time is exactly what separates the classes that currently overlap in the projection.

4

Evaluate the coach with families

Measure whether the cue is understood and acted on, which no figure in this deck can tell us.

One skeleton, one pose bank, one honest set of caveats

Consistent by construction

The notebook figures, the 46 illustrations and the live coach all describe the same body using the same connection list.

Reviewed by eye, not just by code

The coordinate inversion was only ever going to be caught by looking, and it changed how every later figure is checked.

Careful with the claims

52.3% of variance is an indication of separability; 20 to 24 degree tolerances are judgement, not clinical thresholds.

A picture that is consistent with the data is useful. A picture that is mistaken for evidence is not.

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